Stage QA financial transaction tables — offshore user schema workaround

Applies to: Prime **V2** and **V3** Java test automation using `$$QA_SCHEMA$$` for `QA_FIN_TXN` / `QA_FIN_INSTRUCTION` (and related patterns)

1. Overview

Transaction-related tests insert and read **QA-only** financial tables. In Stage, those SQL statements are often built with a **schema prefix** (historically shared schemas such as **UII0** / **UIC**). Many offshore accounts **do not have INSERT/UPDATE** on that shared schema, so tests fail with Oracle permission errors or inconsistent behavior. **DBA access cannot be assumed** to expand shared-schema privileges for every user.

Approved workaround: Create **`QA_FIN_TXN`** and **`QA_FIN_INSTRUCTION`** in **your own Oracle user schema**, then configure automation so `$$QA_SCHEMA$$` expands to **`YOUR_USER.`** (not `UII0.`) on Stage.

2. Root cause

2.1 Why inserts fail against the shared schema

1. **Privilege gap:** Offshore users may have general Stage access but **not** `INSERT`/`UPDATE` on objects in **UII0** (or equivalent shared QA schema).
2. **Wrong object target:** SQL generated as `INSERT INTO UII0.QA_FIN_INSTRUCTION ...` runs as **your** session; without privileges, Oracle returns **`ORA-01031: insufficient privileges`** (or similar).
3. **Trigger / sequence mismatch:** If tables exist in the shared schema but **triggers or sequences** differ from the Pramod DDL, you can see **`ORA-01400: cannot insert NULL`** into `SEQ_QA_FIN_TXN_ID` or related columns when the **BEFORE INSERT** trigger does not populate keys as expected (see sample errors in this folder).

2.2 How schema resolution works in automation

- SQL text in the framework uses the placeholder **`$$QA_SCHEMA$$`** where a schema prefix is required on Stage.
- In **`BaseClass.performQueryReplacements`** (near the end of the method), that placeholder is replaced before the statement runs.
- **Current behavior (before fix):** On Stage, `$$QA_SCHEMA$$` is replaced with a **fixed** shared schema prefix (e.g. **`UII0.`**), so all users hit the same shared objects.

Files to update (Prime automation repo — same logic, two locations):

Stream	File path
V2	`prime/source/com/cs529/qa/prime/core/BaseClass.java`
V3	`prime/prime-core/src/main/java/core/test/BaseClass.java`

> ***Note:*** *This knowledge base does not contain `BaseClass.java`. Confirm the exact snippet in your branch; replace only the $$QA_SCHEMA$$ block as shown in §5.*

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3. Workaround steps (offshore engineer checklist)

Step 1 — Obtain your Oracle Stage username

Use the same user you use in **SQL Developer** (or PuTTY/sqlplus) for Stage 1. Example shape: `SPATIAL_SPAGHETTI` (uppercase is typical in Oracle metadata).

Step 2 — Prepare the DDL script

1. Open `Create QA_FIN_TXN tables DDL.sql attached below.`
2. **Edit the `PUBLIC SYNONYM` lines** so they point to **your** schema, not another user's. The checked-in sample references **`SPATIL`** — replace with **your** username, e.g.:

```
CREATE OR REPLACE PUBLIC SYNONYM QA_FIN_TXN FOR YOUR_USER.QA_FIN_TXN;
CREATE OR REPLACE PUBLIC SYNONYM QA_FIN_INSTRUCTION FOR YOUR_USER.QA_FIN_INSTRUCTION;
```

Create QA_FIN_TXN tables DDL.sql

1. Run the script **connected as `YOUR_USER`** so `CREATE TABLE` creates objects in **your** schema.

> ***Important:*** *If your organization restricts* ***public synonyms*** *or* ***`GRANT ... TO PUBLIC`****, coordinate with DBA; the intent remains:* ***objects live in your schema*** *and your session can `INSERT/SELECT/UPDATE` them.*

Step 3 — Execute the DDL on Stage 1

1. Connect to **Stage 1** as **your** user.
2. Run the full script in **SQL Developer** (Worksheet Run Script), or use **sqlplus** / PuTTY if SQL Developer blocks multi-statement scripts.

If SQL Developer will not run the script: Follow the PuTTY / sqlplus steps in **`How to update data using SQL.docx`** (attach that doc to this Confluence page or link from team SharePoint when available).

Step 4 — Align automation with your schema

1. Apply the **code change** in §5 to **both** V2 and V3 `BaseClass.java` (same method: `performQueryReplacements`).
2. Pass your schema into the JVM (recommended) — see §5 **After** example using **`-Dqa.oracle.schema=YOUR_USER`**.
3. Re-run a failing scenario (e.g. `MemberSingleContribution.feature`).

Step 5 — (Optional) Service ticket path

If policy requires **shared-schema** access instead of user objects, use `instructions to create ticket.txt` in this folder as a **template** for an AGS /access ticket. Historical example: `[AGS #500647] Stage DB Schema_ Access.eml`. **This workaround avoids depending on that approval** for daily automation.

4. Code changes (`performQueryReplacements`)

4.1 Location

- **V2:** `prime/source/com/cs529/qa/prime/core/BaseClass.java` method **`performQueryReplacements`** (near the end).
- **V3:** `prime/prime-core/src/main/java/core/test/BaseClass.java` same method name.

4.2 Before (current pattern)

```
if(query.contains("$$QA_SCHEMA$$")) {
    query = isStageEnvironment() ? query.replace("$$QA_SCHEMA$$", "UII0.") : query.replace("$$QA_SCHEMA$$", "");
}
```

This forces every engineer to use the **UII0** (shared) schema on Stage.

4.3 After (user schema via JVM property)

Use a **dedicated system property** so each offshore engineer sets **their** Oracle user **without** editing code.

```java
if (query.contains("$$QA_SCHEMA$$")) {
    if (isStageEnvironment()) {
        String qaSchema = System.getProperty("qa.oracle.schema", "").trim();
        if (qaSchema.isEmpty()) {
            throw new IllegalStateException(
                "Stage automation requires -Dqa.oracle.schema=<ORACLE_USERNAME> (your Stage user, e.g.
SPATIAL_SPAGHETTI)");
        }
        String prefix = qaSchema.toUpperCase().endsWith(".") ? qaSchema.toUpperCase() : qaSchema.toUpperCase() + "
.";
        query = query.replace("$$QA_SCHEMA$$", prefix);
    } else {
        query = query.replace("$$QA_SCHEMA$$", "");
    }
}
```

Run configuration example (IDE or Maven):

```
-Dqa.oracle.schema=SPATIAL_SPAGHETTI
```

Team lead / US users with shared-schema access can set `-Dqa.oracle.schema=UII0` if they intentionally use the shared objects.

> *If your team prefers a **config file** or **environment variable** instead of `-D`, implement the same string in one place only—keep behavior identical: Stage must resolve to **`SCHEMA.`** or empty non-Stage.*

5. Validation

1. **DB:** As your user, `SELECT COUNT(*) FROM QA_FIN_TXN;` and `QA_FIN_INSTRUCTION` succeed (may be zero rows).
2. **Automation:** Run one previously failing transaction test (e.g. contribution flow). No `ORA-01031` on `UII0.QA_FIN_*`.
3. **Placeholder:** Breakpoint or log the SQL **after** `performQueryReplacements`; confirm $$QA_SCHEMA$$ became **`YOUR_USER.`** on Stage.
4. **ORA-01400:** If NULL PK/FK persists, confirm **triggers** `TRG_QA_FIN_TXN` and `TRG_QA_FIN_INSTRUCTION` exist on **your** tables and that inserts follow order: parent **`QA_FIN_TXN`** then child **`QA_FIN_INSTRUCTION`** where applicable.

6. Sample errors (from team)

Dinesh — `ORA-01400` inserting into `UII0.QA_FIN_INSTRUCTION` (`SEQ_QA_FIN_TXN_ID`): see `Error reported by Dinesh.txt`.

Venkatesh — same symptom on `MemberSingleContribution.feature`: see `Error reported by Venkatesh.txt`.